AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 11 · WORKBOOK

Failure budgets and service objectives

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete service-objective and failure-budget record.

Keep interval identity,eligibility,success rules,failed IDs,exclusions and exact budget arithmetic together.

Keep one habit: Recall → W1–W3/W5 → independent objective → project interval report → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Evaluation pipelines. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Authentication and authorisation. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- **Service objective**s define measured reliability over a stated interval. A **failure budget** is meaningful only after success and **eligible request**s are defined.

- For a fictional objective at least 98% of 200 **eligible request**s must pass all contract checks. Allowed failures are 2%*200=4. Count every eligible failure once; report excluded requests separately.

Evaluation pipelines

> **My remembered idea, example and correction**

Testing model-dependent behaviour

> **My remembered idea, example and correction**

Authentication and authorisation

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main constructed W-A window:200 eligible user requests, 3 internal health checks excluded by predeclared policy. Success requires all supplied schema, support-or-correct-refusal and timing flags.5 requests fail though 7 individual checks fail; count each request once. Success195/200=97.5%. Target≥98% allows at most 4 failures; 5 exceeds by1, so this window misses.

  Correct refusal can be success under the declared contract. Exact integer maximum is floor((100–target_percent)×eligible/100); 98% over 12 allows0 failures. Separate picture original 100 requests has 97 success/3 fail, allowance 2, misses by1. A counterfactual copy 98/2 meets at equality; preserve original 97/3. A real retry does not silently erase a previous failed request.

  Zero eligible denominator has no empirical rate. Synthetic records do not establish actual timing, live reliability, zero critical incidents or future performance.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main constructed W-A window:200 eligible user requests, 3 internal health checks excluded by predeclared policy. Success requires all supplied schema, support-or-correct-refusal and timing flags.5 requests fail though 7 individual checks fail; count each request once. Success195/200=97.5%. Target≥98% allows at most 4 failures; 5 exceeds by1, so this window misses.

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  Zero eligible denominator has no empirical rate. Synthetic records do not establish actual timing, live reliability, zero critical incidents or future performance.

- In the interval 200 requests are eligible and 5 fail. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main constructed W-A window:200 eligible user requests, 3 internal health checks excluded by predeclared policy. Success requires all supplied schema, support-or-correct-refusal and timing flags.5 requests fail though 7 individual checks fail; count each request once. Success195/200=97.5%. Target≥98% allows at most 4 failures; 5 exceeds by1, so this window misses.

  Correct refusal can be success under the declared contract. Exact integer maximum is floor((100–target_percent)×eligible/100); 98% over 12 allows0 failures. Separate picture original 100 requests has 97 success/3 fail, allowance 2, misses by1. A counterfactual copy 98/2 meets at equality; preserve original 97/3. A real retry does not silently erase a previous failed request.

  Zero eligible denominator has no empirical rate. Synthetic records do not establish actual timing, live reliability, zero critical incidents or future performance.

- A report omits this qualification: Meeting one interval objective does not establish every future interval or zero critical failures.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main constructed W-A window:200 eligible user requests, 3 internal health checks excluded by predeclared policy. Success requires all supplied schema, support-or-correct-refusal and timing flags.5 requests fail though 7 individual checks fail; count each request once. Success195/200=97.5%. Target≥98% allows at most 4 failures; 5 exceeds by1, so this window misses.

  Correct refusal can be success under the declared contract. Exact integer maximum is floor((100–target_percent)×eligible/100); 98% over 12 allows0 failures. Separate picture original 100 requests has 97 success/3 fail, allowance 2, misses by1. A counterfactual copy 98/2 meets at equality; preserve original 97/3. A real retry does not silently erase a previous failed request.

  Zero eligible denominator has no empirical rate. Synthetic records do not establish actual timing, live reliability, zero critical incidents or future performance.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Objective 95% over 80 **eligible request**s. Five fail. Compute allowed failures and observed success.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled **failure budget**s and **service objective**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **failure budget**s and **service objective**s.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Failure budget**s and **service objective**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Failure budget**s and **service objective**s. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Objective 95% over 80 **eligible request**s. Five fail. Compute allowed failures and observed success.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

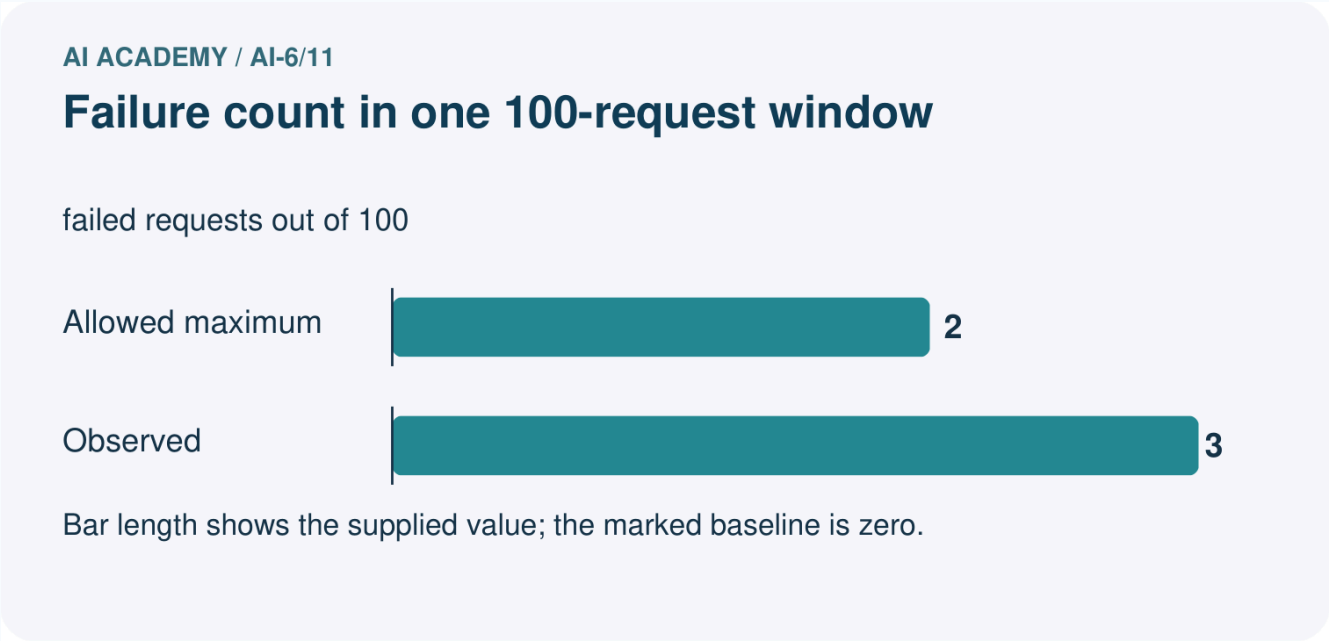
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A reliability allowance. A practice service requires at least 98 successes in each window of 100 **eligible request**s.
- Current window has 97 successes and 3 failures. Success includes a correct refusal within the agreed time limit.
- Calculate the maximum allowed failures among 100 requests. Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-6/11

Failure budgets and service objectives

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Define success

Use the same predeclared correctness and timing criteria.



2. Compute allowance

The target permits at most 2 failures in 100 requests.



3. Compare record

Three failures misses the target by one request.

CHANGE ONE THING

Repair one failed request into a success in this window.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A reliability allowance. A practice service requires at least 98 successes in each window of 100 **eligible request**s.
- Which labelled step matches this detail? Three failures misses the target by one request.
- 1. Define success
- 2. Compute allowance
- 3. Compare record
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare record Three failures misses the target by one request. An objective needs a measurement window, denominator and success definition.

An honest refusal can be success under an agreed contract.

Changed-case question and optional hint

- Changed-case practice: Counterfactual: change one failed outcome to a success in a separate copy of this 100-request window. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. 98 successes and 2 failures meet the stated window target.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: 98 successes and 2 failures meet the stated window target. The counterfactual copy changes one outcome while its denominator stays 100; preserve the original 97-success/3-failure window.

Repair and retain the record

Review Evaluation pipelines (ai-6/10). Goal: Explain the **mechanism** and terminology of evaluation pipelines. Short practice: Calculate A's and B's correct rates on all four cases.

Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.. Return to this week and explain what you changed.

Review Testing model-dependent behaviour (ai-6/8). Goal: Explain the **mechanism** and terminology of testing model-dependent behaviour.

Short practice: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?. Return to this week and explain what you changed.

Review Authentication and authorisation (ai-6/5). Goal: Explain the **mechanism** and terminology of authentication and authorisation. Short practice: Explain why Jo's valid volunteer login does not permit editing C.

Which trusted role change allows the action, and why is a role written in the request insufficient?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Failure budgets and service objectives: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

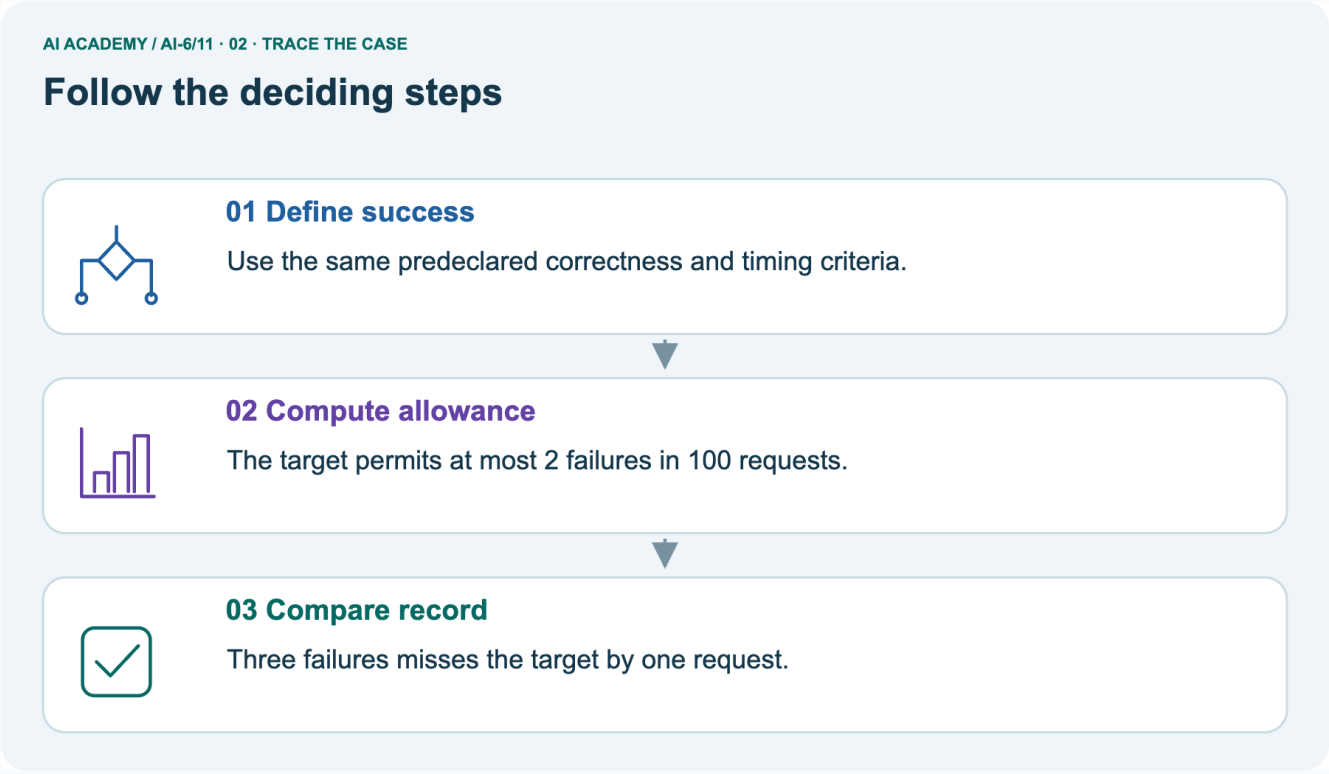
A reliability allowance. A practice service requires at least 98 successes in each window of 100 **eligible request**s. Supplied facts: Current window has 97 successes and 3 failures.

Success includes a correct refusal within the agreed time limit.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



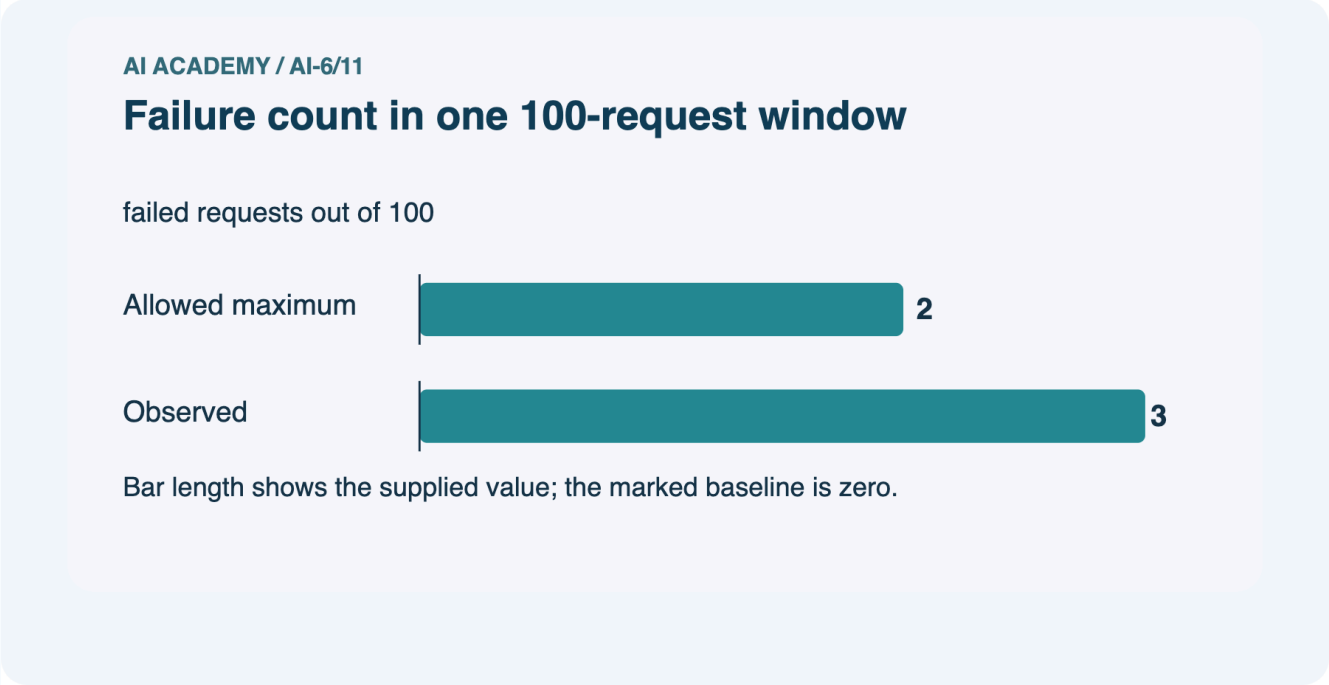
Follow the labelled picture with the matching explanation. Open the vector diagram

Define success: Use the same predeclared correctness and timing criteria. → Compute allowance: The target permits at most 2 failures in 100 requests. → Compare record: Three failures misses the target by one request.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Failure count in one 100-request window



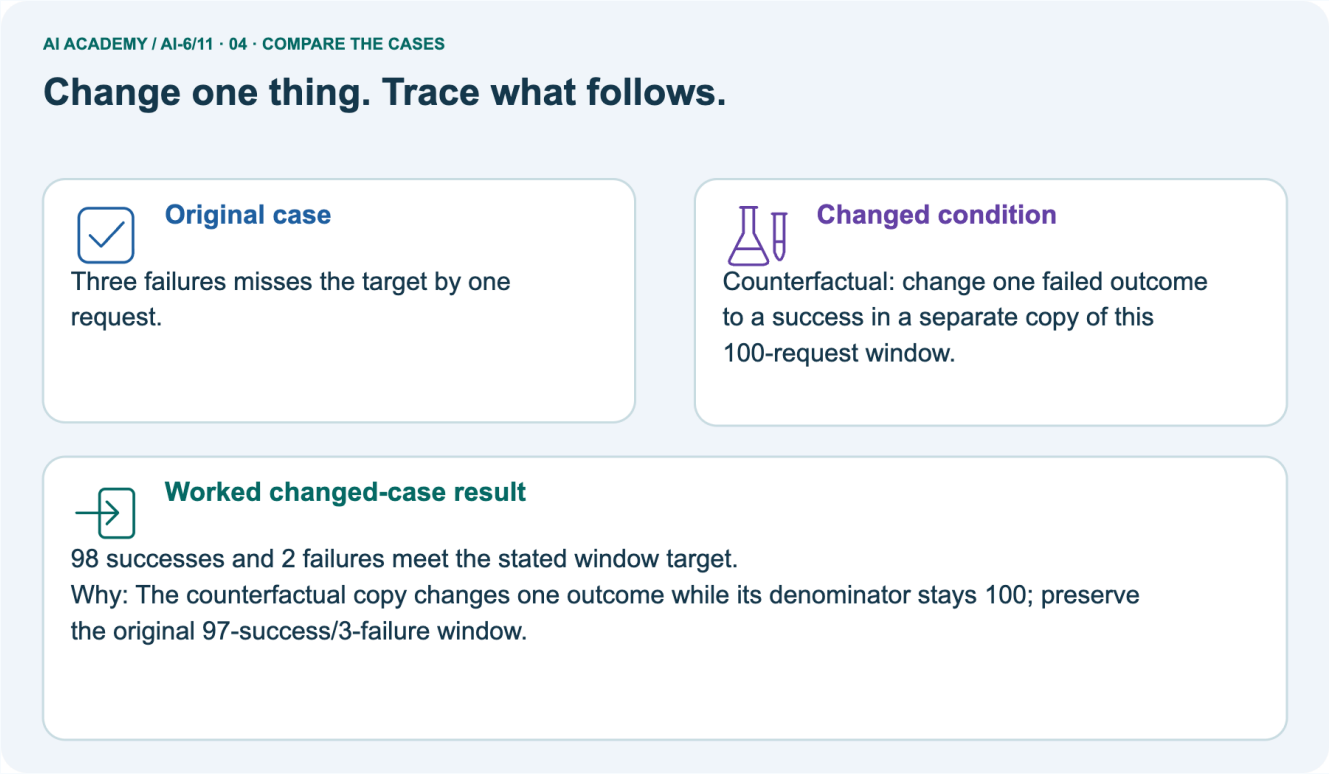
Follow the labelled picture with the matching explanation. Open the vector diagram

Failure count in one 100-request window. Units: failed requests out of 100. Allowed maximum: 2. Observed: 3.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



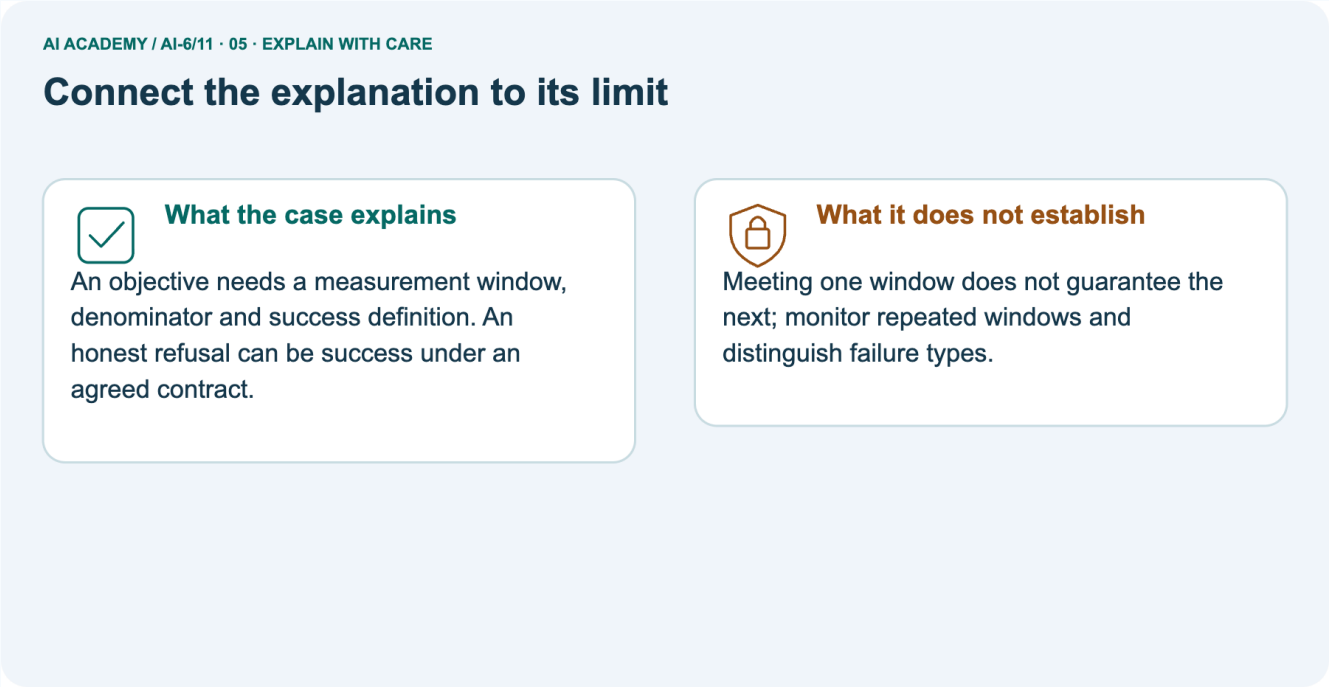
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Three failures misses the target by one request. Changed condition: Counterfactual: change one failed outcome to a success in a separate copy of this 100-request window. Expected result: 98 successes and 2 failures meet the stated window target. Reason: The counterfactual copy changes one outcome while its denominator stays 100; preserve the original 97-success/3-failure window.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

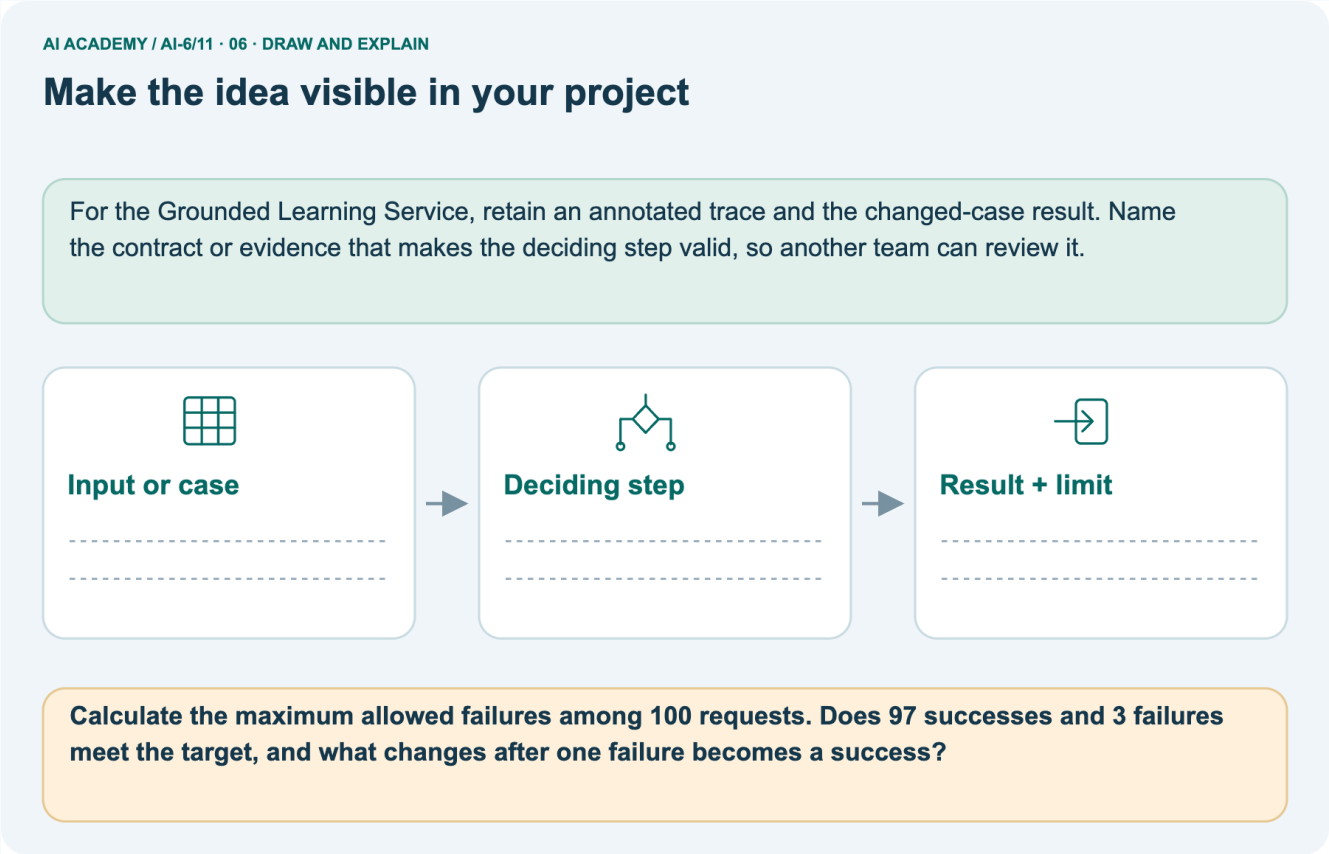
An objective needs a measurement window, denominator and success definition. An honest refusal can be success under an agreed contract.

Limit: Meeting one window does not guarantee the next; monitor repeated windows and distinguish failure types.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Failure budgets and service objectives: the case

02. Follow the deciding steps

03. Failure count in one 100-request window

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.